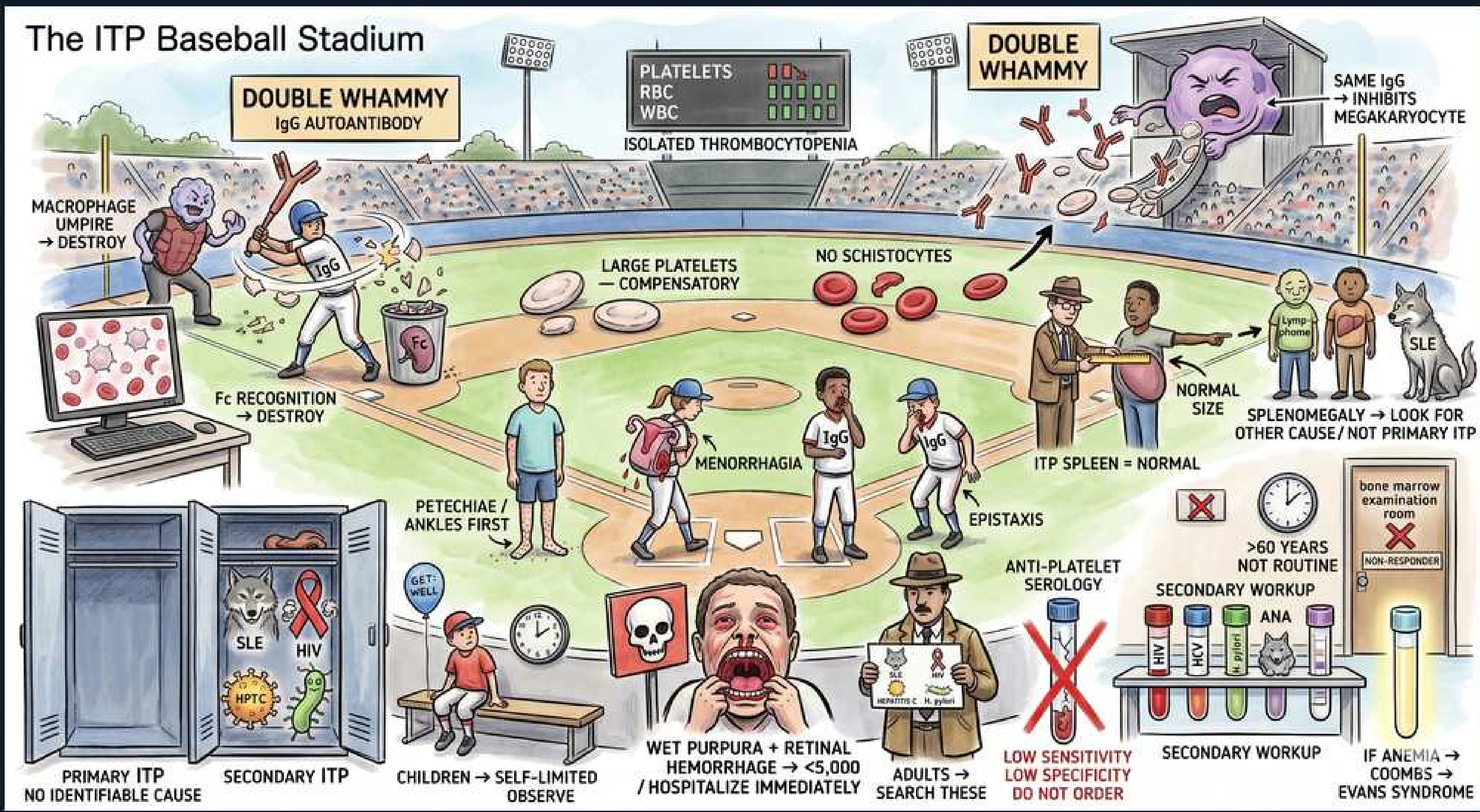
Platelet disorders — ITP (The ITP Baseball Stadium)



1. IgG autoantibody

WHAT YOU SEE

A 'DOUBLE WHAMMY — IgG AUTOANTIBODY' banner over a batter swinging an IgG-tagged bat — the same antibody both destroys platelets and poisons their production.

WHAT IT ENCODES

ITP (immune thrombocytopenia) is autoantibody-mediated: IgG directed against platelet surface glycoproteins, most classically GPIIb/IIIa and GPIb/IX, opsonizes circulating platelets. This is a type II hypersensitivity reaction. The 'double whammy' is the key concept — the same antibody both tags mature platelets for destruction AND poisons production, so counts fall faster than a pure-destruction model would predict.

BOARD TRAP

A stem describing thrombocytopenia plus a long PTT/PT or low fibrinogen is NOT ITP — that is consumptive coagulopathy (DIC). In ITP the coagulation studies (PT, aPTT, fibrinogen) are completely NORMAL because only platelets are targeted; the wrong answer is to attribute isolated antibody-mediated platelet loss to a clotting-cascade problem.

---

2. Isolated thrombocytopenia

WHAT YOU SEE

The stadium scoreboard reading PLATELETS down but RBC and WBC normal, labeled 'ISOLATED THROMBOCYTOPENIA' — only the platelet line is low.

WHAT IT ENCODES

The defining lab finding is ISOLATED thrombocytopenia: low platelets with a completely normal hemoglobin and white count, and a normal smear apart from low platelet number (plus some large platelets). There is no anemia, no leukopenia, no blasts. Typical 'treat' threshold is <30,000; spontaneous serious bleeding risk climbs steeply below 10,000–20,000.

BOARD TRAP

If the CBC shows a second or third cytopenia (anemia, leukopenia, or pancytopenia), do NOT call it ITP — that pattern points to a bone-marrow process (aplastic anemia, MDS, leukemia, marrow infiltration) or a microangiopathy (TTP/HUS). Picking 'ITP' when two cell lines are down is the classic distractor; ITP touches platelets only.

---

3. Megakaryocyte inhibition

WHAT YOU SEE

A megakaryocyte in the upper stands being hit by the 'SAME IgG -> INHIBITS MEGAKARYOCYTE' — the second whammy, antibody blocking marrow production too.

WHAT IT ENCODES

Second hit of the double whammy: the same anti-GPIIb/IIIa IgG also binds megakaryocytes in the marrow and impairs platelet production and maturation. So ITP is not pure peripheral destruction — there is also relative underproduction. This is the rationale for TPO receptor agonists (eltrombopag, romiplostim), which override the production block.

BOARD TRAP

Students assume marrow in ITP shows DECREASED megakaryocytes; the wrong answer. Marrow (when done) shows NORMAL-to-INCREASED megakaryocytes because the marrow is trying to compensate for peripheral destruction. A stem describing absent/reduced megakaryocytes suggests aplastic anemia or marrow failure, not ITP.

---

4. Macrophage destruction

WHAT YOU SEE

A macrophage umpire devouring a platelet, tagged 'DESTROY' — the splenic macrophage that phagocytoses IgG-coated platelets via its Fc receptor.

WHAT IT ENCODES

IgG-coated platelets are cleared by splenic macrophages, which recognize the bound Fc portion of IgG through their Fcγ receptors and phagocytose the platelet — the spleen is both the main site of antibody production and the main graveyard. This Fc-mediated clearance is exactly what IVIG (Fc-receptor blockade) and splenectomy (remove the graveyard) target.

BOARD TRAP

Don't confuse the splenic Fc-receptor clearance of ITP with complement-mediated intravascular lysis. ITP platelet destruction is predominantly extravascular (reticuloendothelial/splenic phagocytosis), not complement-driven hemolysis; a stem stressing complement and intravascular destruction is steering you toward a different mechanism (e.g., PNH, transfusion reactions).

---

5. Large platelets

WHAT YOU SEE

Oversized platelets on the field labeled 'LARGE -- COMPENSATORY' with 'NO SCHISTOCYTES' nearby — big young platelets from a revved-up marrow and a smear free of fragments.

WHAT IT ENCODES

The peripheral smear shows LARGE platelets (megathrombocytes) reflecting young platelets released early from a hyper-stimulated marrow, and the smear is otherwise unremarkable. Critically, there are NO schistocytes. Large platelets + normal RBC morphology + isolated thrombocytopenia is the ITP smear.

BOARD TRAP

Seeing schistocytes (fragmented helmet/triangle RBCs) on the smear takes you OUT of ITP and INTO a microangiopathic hemolytic anemia — TTP, HUS, or DIC. The trap answer is to keep calling it ITP after the stem mentions fragmented red cells; schistocytes mean mechanical RBC shearing, which ITP does not cause.

---

6. Spleen normal size

WHAT YOU SEE

A normal-sized spleen captioned 'ITP SPLEEN = NORMAL; SPLENOMEGALY -> LOOK FOR OTHER CAUSE' — primary ITP keeps a normal spleen, so a big spleen points elsewhere.

WHAT IT ENCODES

In primary ITP the spleen is NORMAL in size — it is hyperfunctional but not enlarged. Palpable splenomegaly is a red flag that argues against primary ITP and should redirect you toward a secondary cause: lymphoproliferative disease (CLL, lymphoma), liver disease with portal hypertension/hypersplenism, or infection.

BOARD TRAP

A stem with thrombocytopenia AND a palpable spleen should NOT be answered 'primary ITP.' Splenomegaly + cytopenias points to hypersplenism (sequestration), CLL, lymphoma, or portal hypertension. Choosing primary ITP in the face of a big spleen is the wrong move — primary ITP keeps a normal-size spleen.

---

7. Petechiae / mucosal bleed

WHAT YOU SEE

Players showing petechiae starting at the ankles plus menorrhagia and epistaxis — the mucocutaneous, dependent-first bleeding of a platelet problem.

WHAT IT ENCODES

ITP bleeding is the classic platelet-type pattern: MUCOCUTANEOUS — petechiae (start dependent, at the ankles), purpura, epistaxis, gum bleeding, and menorrhagia. Bleeding is superficial and immediate. The dreaded but rare complication is intracranial hemorrhage, mostly at counts <10,000–20,000.

BOARD TRAP

Don't confuse the platelet-type bleeding of ITP (petechiae, mucosal, immediate, superficial) with the coagulation-factor pattern (deep hematomas, hemarthroses, delayed rebleeding) seen in hemophilia. A stem with bleeding into joints and muscles is a clotting-factor deficiency, NOT ITP; petechiae/mucosal bleeding is the platelet signature.

8. Primary vs secondary

WHAT YOU SEE

A locker room with SLE, HIV, and HCV lockers labeled 'SECONDARY ITP' — the underlying causes to screen for behind ~20% of ITP.

WHAT IT ENCODES

Roughly 20% of ITP is SECONDARY — driven by an underlying disorder you must screen for: HIV, HCV (and H. pylori in some settings), SLE and other autoimmune disease, CLL/lymphoma, and drugs. Board reflex: a new ITP work-up in an adult always includes HIV and HCV testing. Treating the underlying cause (e.g., HCV antivirals, HIV ART) can resolve the thrombocytopenia.

BOARD TRAP

Forgetting to test for HIV and HCV in adult new-onset 'ITP' is the classic omission the boards punish. The wrong answer is to start immunosuppression without screening — missing a treatable secondary cause (HIV/HCV/SLE/CLL) is a management error even when steroids would 'work.'

9. Children self-limited

WHAT YOU SEE

A child beside a 'GET WELL' clock marked 'CHILDREN -> SELF-LIMITED, OBSERVE' — pediatric ITP follows a virus and resolves on its own.

WHAT IT ENCODES

Childhood ITP is typically ACUTE, follows a viral infection or vaccination by 1–4 weeks, and is SELF-LIMITED — most kids recover spontaneously within weeks to a few months, so observation is the default if there is no significant bleeding. This contrasts sharply with adult ITP, which is usually insidious and chronic.

BOARD TRAP

The trap is to immediately start steroids/IVIG in a well child with isolated thrombocytopenia and only minor cutaneous findings. Pediatric ITP without active bleeding is OBSERVED — reflexively treating by the number alone is wrong; intervene for active or severe (wet) bleeding, not for the platelet count in isolation.

10. Wet purpura — admit

WHAT YOU SEE

A face with bloody oral blisters and 'WET PURPURA + RETINAL HEMORRHAGE <5,000 -> HOSPITALIZE IMMEDIATELY' — mucosal/retinal bleeding marks high ICH risk needing admission.

WHAT IT ENCODES

'Wet purpura' = mucosal/oral blood blisters, plus signs like retinal hemorrhage, GI/GU bleeding, or any count <5,000–10,000 — these mark HIGH risk of life-threatening bleed (including ICH) and warrant HOSPITALIZATION and urgent treatment (high-dose steroids + IVIG, platelets if life-threatening). 'Dry purpura' (skin-only petechiae) is lower risk.

BOARD TRAP

Do NOT discharge or merely observe a patient with WET purpura or active mucosal/retinal bleeding regardless of how stable they look — the wrong answer is outpatient follow-up. Wet purpura is the bleeding phenotype that drives the decision to admit and treat urgently, not the absolute platelet number alone.

11. Diagnosis of exclusion

WHAT YOU SEE

A trench-coated detective captioned 'ADULTS -> SEARCH THESE' — ITP is a diagnosis of exclusion, made only after hunting down mimics.

WHAT IT ENCODES

ITP is a DIAGNOSIS OF EXCLUSION — there is no confirmatory test. You make it by demonstrating isolated thrombocytopenia, a compatible smear, and ruling out mimics: pseudothrombocytopenia (EDTA clumping — recheck citrate tube), drugs (heparin/HIT, quinine, sulfa), infections (HIV, HCV), TTP/DIC, marrow disease, and hypersplenism.

BOARD TRAP

Pseudothrombocytopenia is the favorite gotcha: an asymptomatic patient with an isolated low platelet count and platelet CLUMPING on the smear has EDTA-induced clumping, NOT ITP. The right next step is to REDRAW in a citrate (blue-top) tube, not to start steroids; treating lab artifact as ITP is the error.

12. Anti-platelet Ab not useful

WHAT YOU SEE

Anti-platelet serology tubes slashed with a red X, 'LOW SENSITIVITY / LOW SPECIFICITY -- DO NOT ORDER' — the antibody assay that neither rules ITP in nor out.

WHAT IT ENCODES

Anti-platelet antibody testing has POOR sensitivity and specificity and does NOT establish or exclude ITP — it is explicitly not recommended in routine work-up. Diagnosis remains clinical/exclusionary. Money is better spent on HIV, HCV, and a careful smear review.

BOARD TRAP

Selecting 'anti-platelet antibody assay' as the confirmatory/next diagnostic test is the planted wrong answer. A negative assay does not rule out ITP and a positive one does not rule it in — ordering it changes nothing. The boards want you to recognize it as low-yield and choose exclusion of secondary causes instead.

13. Marrow biopsy not routine

WHAT YOU SEE

A 'BONE MARROW EXAMINATION ROOM' door crossed out with '>60 YEARS NOT ROUTINE' — marrow biopsy is not needed for classic ITP, reserved for atypical/older cases.

WHAT IT ENCODES

Bone marrow biopsy is NOT routinely required to diagnose ITP. Reserve it for atypical features — additional cytopenias, abnormal smear cells/blasts, poor response to first-line therapy, or older patients (commonly cited >60) where MDS is a real mimic. When done in ITP, it shows normal/increased megakaryocytes.

BOARD TRAP

Reflexively ordering a bone marrow biopsy in a young patient with classic isolated thrombocytopenia is the over-testing trap. Conversely, SKIPPING marrow in an older adult (>60) with atypical features where MDS must be excluded is the under-testing trap. The discriminator is age plus atypical features, not ITP itself.

14. Evans syndrome

WHAT YOU SEE

A panel reading 'IF ANEMIA + COOMBS -> EVANS SYNDROME' — ITP plus Coombs-positive autoimmune hemolytic anemia is Evans, not TTP.

WHAT IT ENCODES

Evans syndrome = autoimmune thrombocytopenia (ITP) PLUS autoimmune hemolytic anemia, with a POSITIVE direct Coombs (DAT) and signs of hemolysis (low haptoglobin, high LDH, high indirect bilirubin, spherocytes). It is often associated with SLE, CLL, or immunodeficiency and tends to be more refractory than isolated ITP.

BOARD TRAP

When thrombocytopenia is accompanied by anemia with a POSITIVE Coombs and spherocytes, the answer is Evans syndrome (autoimmune), NOT TTP. The key discriminator: TTP/HUS hemolysis is mechanical and Coombs-NEGATIVE with schistocytes, whereas Evans is Coombs-POSITIVE with spherocytes. Calling a Coombs-positive hemolytic anemia 'TTP' is the trap.